Large prime degree isogenies in isogeny_algos (2026-10-16)

Question. Make isogenies of large prime degree ℓ faster in `isogeny_algos`. That covers kernel → codomain and point images for Montgomery and short Weierstrass curves at $\ell = 101 \ldots 250007$ over 256- and 511-bit primes.

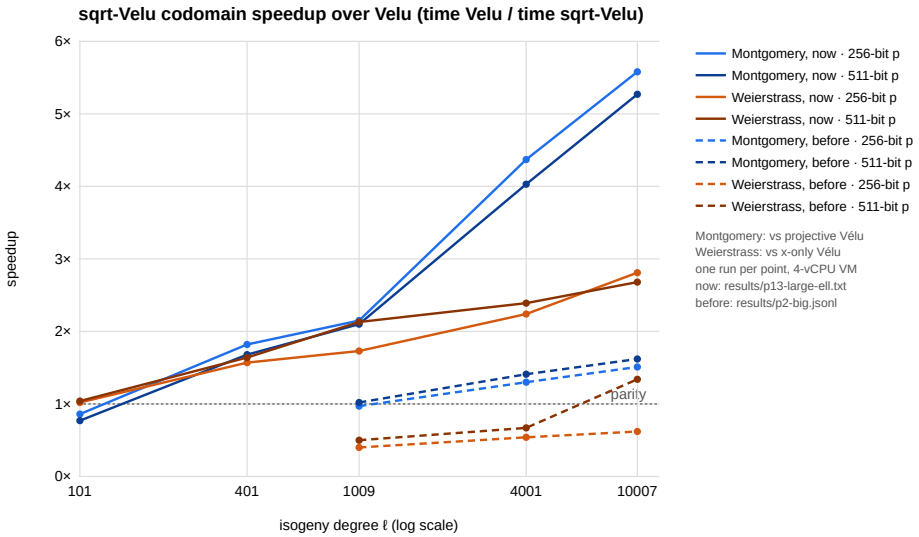
Status of each statement. Speedups are *measured* (one run per setting on a shared 4-vCPU VM; expect ±10–20 % noise). Correctness is *verified* by exact equality with Vélu in the crate's tests (110/110 pass; 130/130 after merging current `main`). The algebraic identities are *derived* below and checked by those tests. No claim here concerns ECDLP.

Method

√élu splits the kernel half $S = (I \pm J) \cup K$ (Bernstein–De Feo–Leroux–Smith 2020). For a pair (i, j), the x-coordinates $x\pm = x(P_i \pm P_j)$ are the roots of a biquadratic $F0\,T^2 + F1\,T + F2$ in x_i, x_j.

1. **Montgomery codomain, pair by pair.** The codomain needs $h_S(1)$ and $h_S(-1)$, up to a common factor. At $\alpha = \pm 1$ each factor $c2\,Z^2 + c1\,Z + c0$ of E_J has $c0 = c2$, so `E_J(Z) = Z^b ∏_j (c2_j W + c1_j)` with `W = Z + 1/Z`. Hence $h_S(\pm 1)$ is a product over pairs of `(c2_j w_i + c1_j)` with `w_i = x_i + 1/x_i`. With `s_j = c1_j / c2_j` (one batch inversion) that is `(∏ c2_j)^b' · ∏_{i,j} (w_i + s_j)`: **one multiplication per pair**, about $\ell/2$ for the codomain, against about 8 per kernel point for projective Vélu.
2. **Montgomery images.** For each pair, $q_j(x_i)$ and its reverse share $c1 \cdot x_i$: five multiplications per pair, so 2.5 per kernel point against 4 for Vélu. The projective-input variant homogenises in (X, Z) and needs no inversion.
3. **Weierstrass.** Each pair gives $x_+ + x_- = 2((x_i x_j + A)(x_i + x_j) + 2B)/(x_i - x_j)^2$ and $x_+x_- = ((x_i x_j - A)^2 - 4B(x_i + x_j))/(x_i - x_j)^2$, with one batch inversion for all pairs. From these come the power sums p_1, p_2, p_3 (codomain), and for images the sums $\Sigma\, 1/(\alpha - x)^k$, $k = 1, 2, 3$.
4. **$\tilde{O}(\sqrt{\ell})$ path kept, opt-in.** It uses a Bernstein scaled remainder tree (`poly::EvalTree`) and the W-product. It is 2.1–2.8× faster than the previous tree code, but slower than pair-by-pair products at every measured size up to $\ell = 250007$. The cause is the lazy multiply-accumulate under the polynomial products, which costs 0.85–1.1 field multiplications per term.
5. **x-only Vélu.** Differential addition specialised to the affine base point, small multiples computed by additions, and cheaper image sums: 15–25 % faster.

Results



sqrt-Velu codomain speedup over Velu

ℓ	Montgomery codomain 256 / 511 bits	one image 256 / 511 bits	Weierstrass codomain 256 / 511 bits	image 256 / 511 bits
101	0.86× / 0.77×	1.16× / 1.19×	1.02× / 1.04×	1.32× / 1.33×
401	1.82× / 1.68×	1.36× / 1.36×	1.57× / 1.64×	1.60× / 1.56×
1009	2.15× / 2.10×	1.49× / 1.43×	1.73× / 2.13×	1.60× / 1.65×
4001	4.37× / 4.03×	1.48× / 1.49×	2.24× / 2.39×	1.74× / 1.65×
10007	5.58× / 5.27×	1.51× / 1.27×	2.81× / 2.68×	1.97× / 1.88×

The comparators are projective Vélu for Montgomery curves and x-only Vélu for Weierstrass curves. The Weierstrass comparator is itself 15–25 % faster than before this change.

Before this change, the recorded run `results/p2-big.jsonl` gives:

- **Montgomery √élu codomain:** 1.51× (256 bits) and 1.62× (511 bits) at ℓ = 10007, slower than Vélu below ℓ ≈ 1000.
- **Weierstrass √élu:** slower than x-only Vélu at 256 bits at every ℓ measured. At 256 bits, ℓ = 10007, √élu went from 5.13 ms to 1.16 ms.

Negative results.

- **Pair-by-pair vs remainder tree (Montgomery codomain).** The tree path takes 1.7–2.8× as long as the pair-by-pair products for ℓ = 10007 … 250007.
- **Images.** The tree path also takes 1.07–1.7× as long as pair by pair at those sizes.
- **Product-scanning multiply-accumulate.** It cost 1.1–1.9 field multiplications per term, so it was dropped.
- **CSIDH-512 with √élu leaves.** Using √élu from ℓ = 151 or 263 gave no change outside the noise: 31–32 ms in alternating runs over 12 keys, with identical outputs. The option `Csidh::sqrt_velu_min` is therefore off by default.

Evidence

- Raw output: `isogeny_algos/results/p13-large-ell.txt`. It is produced by the examples `profile_large_ell`, `profile_weier`, `profile_tree_vs_direct` and `profile_csidh`.
- Tests:
  - `tests/sqrt_velu_mont.rs`: both paths, `new_proj` and `eval_xz` against Vélu on all 74 CSIDH-512 degrees and on $\ell = 1009$ and 10007;
  - `tests/polyfast.rs`: `EvalTree` against Horner;
  - `tests/kernel2.rs` and `tests/bigfield.rs`: Weierstrass √élu codomain and images against x-only Vélu.